

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0704

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

ANALYTICAL PROBLEM SOLVING

Code of Conduct:

U192524291)

I S. BHODMIKA (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Bhodmika

Annexure A

ANALYTICAL PROBLEM SOLVING

1. A university network is assigned the IP address block 192.168.20.0/24 for its internal communication. The network administrator has decided to divide this network into subnets for three different labs using fixed subnet masks. Lab A is assigned a /26 subnet, Lab B is assigned a /27 subnet, and Lab C is assigned a /28 subnet. Based on the given subnet masks, apply subnetting concepts to determine the network address of each lab. Further, identify the valid range of host IP addresses and the broadcast address for each subnet. Finally, calculate the total number of usable host IP addresses available in each lab's subnet.
2. A network administrator has configured two systems with the following details: System 1 has IP address 192.168.1.130 with subnet mask 255.255.255.192, and System 2 has IP address 192.168.1.190 with subnet mask 255.255.255.192. Using the given configuration, apply subnetting techniques to determine the subnet range, network address, and broadcast address for each system. Identify the valid host IP range for both subnets based on the given subnet mask. Calculate whether each IP address falls within its respective subnet range. Also, determine the number of usable host addresses available in each subnet.
3. An organization has been allocated the IP block **172.16.0.0/16** for its internal network. The network administrator has already decided the subnet masks for each department as follows: HR → /25, Sales → /26, IT → /27, and Admin → /28. Using the given subnet masks, apply subnetting techniques to assign suitable subnet addresses for each department. Determine the network address, valid host IP range, and broadcast address for each subnet. Based on the allocation, calculate the number of usable host addresses in each department.

Finally, identify the remaining unused IP address range within the given network block.

4. A company uses the IP address **172.16.10.0/24** and has implemented subnetting using a **/27 subnet mask** for its internal departments. Using the given subnet mask, apply subnetting techniques to calculate the total number of subnets created and the number of usable hosts per subnet. Determine the subnet to which the IP address **172.16.10.78** belongs. Identify the network address and broadcast address for that subnet. Also, determine whether the IP address **172.16.10.95** falls within the same subnet range based on the given configuration.

5. A network uses the Internet Protocol version 6 address **2001:0db8:0000:0000:0000:ff00:0042:8329**. Apply IPv6 addressing rules to compress the given address into its shortest form and then expand the compressed address back to its full notation. Using a prefix length of **/64**, determine the network prefix of the given address. Also, calculate the total number of possible host addresses available within this network based on the given prefix length.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation

IPv4 Subnetting

Given :

IP address : 172.16.10.0/24

Subnet mask : /27

IP addresses : 172.16.10.78 and 172.16.10.95

Step 1 : calculate total number of subnets.

Original prefix = /24

New prefix = /27

Bits borrowed : $27 - 24 = 3$

Total number of subnets = $2^3 = 8$

Step 2 : calculate usable hosts per subnet

Remaining host bits = $32 - 27 = 5$

Total addresses per subnet = $2^5 = 32$

Usable hosts = $32 - 2 = 30$

Step 3 : list the subnets

subnet	network address	host range	Broadcast address
1.	172.16.10.0	1-30	172.16.10.31
2.	172.16.10.32	33-62	172.16.10.63
3.	172.16.10.64	65-94	172.16.10.95

4.	172.16.10.96	97-120	172.16.10.121
5.	172.16.10.128	129-158	172.16.10.159
6.	172.16.10.160	161-190	172.16.10.223
7.	172.16.10.192	193-222	172.16.10.255
8.	172.16.10.224	225-254	

Step 4: Find the subnet for IP address

172.16.10.78.

The IP 172.16.10.78 lies between 65 and 94

Therefore,

Subnet: 172.16.10.64/27

Network Address: 172.16.10.64

Broadcast Address: 172.16.10.95

Usable host range: 172.16.10.65 - 172.16.10.94

Step 5: check whether 172.16.10.95 belongs to the same subnet

Network Address = 172.16.10.64

Broadcast Address = 172.16.10.95

Hence,

172.16.10.95 is the broadcast address of this subnet.

addressing :
given address :

2001: 0db8 : 0000 : 0000 : 0000 : f400 : 0042 : 8329

prefix length = 164

step 1 : compress the IPv6 address.

Rules :

- * Remove leading zeroes in each block
- * Replace one continuous sequence of zero blocks with "::".

compressed address :

2001 : db8 :: f400 : 42 : 8329.

step 2 : Expand Back the full form

Expanded address :

2001 : 0db8 : 0000 : 0000 : 0000 :: f400 : 0042 : 8329

step 3 : determination the network prefix (164)
A 164 prefix means the first 64 bits represent the network.

network prefix :

2001 : 0db8 : 0000 : 0000 :: 164

or in compressed form,

2001 : db8 :: 164.

Step 4: calculate Total host addresses.

IPv6 address length: 128 bits.

prefix length = 64 bits

host bits = $128 - 64 = 64$

Total possible host addresses

$= 2^{64} = 18.4 \text{ quintillion addresses.}$

Final address:

original address:

2001:0db8:0000:0000:0000:ffff:0042:8329

compressed address:

2001:db8::ffff:42:8329

Expanded address:

2001:0db8:0000:0000:0000:ffff:0042:

8329.

Network prefix (164) : 2001:db8::/64

host bits: 64

Total possible host addresses $= 2^{64} =$
18.4 quintillion